Status	Finished
Started	Saturday, 10 October 2026, 8:08 AM
Completed	Saturday, 10 October 2026, 8:18 AM
Duration	9 mins 12 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists
b. It can be applied only if the list is sorted ✓
c. It always starts from the beginning of the list
d. It compares elements sequentially

Question **2**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted
b. The search ends successfully ✓
c. The search continues in the left sublist
d. The search continues in the right sublist

Question **3**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. log n
c. n/2
d. n ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Hash search

b. Linear search ✓

c. Both (1) & (3)

d. Binary search

Question **5**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)

b. O(n log n)

c. O(1)

d. O🥊 ✓

Question **6**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Sort the list

b. Compare the target element with the first element in the list

c. Compare the target element with the middle element in the list ✓

d. Divide the list into two equal parts

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The last element is returned
c. The first element is returned
d. The search is considered unsuccessful ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Deleting elements from a list
b. Inserting elements into a list
c. Determining whether an element is present in a list ✓
d. Sorting elements in a list

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
b. Must use a sorted array
c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
d. There must be a mechanism to access middle element directly

Question **10**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Interpolation search
b. Hash search
c. Linear search ✓
d. Binary search

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Checking each element sequentially ✓
- b. Skipping every second element
- c. Sorting the list before searching
- d. Dividing the list in half repeatedly

Question **12**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O🥊
- c. O(n log n)
- d. O(1)

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By dividing the list into halves
- b. By sorting the list first
- c. By comparing each element in the list sequentially ✓
- d. By using a hash function

Question **14**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n/2
- c. n-1
- d. log n

Question **15**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O(1) ✓
c. O👎
d. O(n log n)